

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ONLINE CODE JUDGE

Database Systems Engineering & Distributed Backend Development (25CS1302E)

ACADEMIC SESSION

Y25 | 2026-2027

Trimester 4 • Section 10

EVALUATION STAGE

Project Review 2

Frontend & 3NF Data Design

TEAM DETAILS

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Problem Statement

System Rationale & Context

Synchronous Execution Bottlenecks: Traditional judges block web servers during code compilation and testing, causing request timeouts and platform failures during contest traffic bursts.

Security Boundaries for Untrusted Code: Running user-submitted code requires hard container isolation with strictly enforced ceilings on CPU runtime, resident memory, and system calls.

Architectural Coupling: Monolithic architectures intertwine problem authoring, submission auditing, and live ICPC score calculations, leading to cascading system failure.

Engineering Objective: Construct a modular, distributed architecture that decouples browser code ingestion from sandboxed worker execution.



System Objectives

Review 2 Scope

1. Monaco Code Workspace

IMPLEMENTED

Integrate a multi-language browser IDE supporting C++, Python 3, Java, and JavaScript with syntax validation, local code caching, and keyboard-first shortcuts.

Evidence: ProblemWorkspace.tsx & MonacoCodeEditor.tsx

2. Problem Authoring & Catalog

IMPLEMENTED

Faceted catalog search across difficulty tiers, topic tags, and live solve status, paired with a CRUD modal for problem setters to input test cases.

Evidence: ProblemCatalog.tsx & CreateProblemModal.tsx

3. Contests & Leaderboard Arena

IMPLEMENTED

Contest tournament scheduling and real-time scoreboards tracking user solves, attempt counts, and cumulative ICPC penalty minutes.

Evidence: ContestsView.tsx & LeaderboardView.tsx

4. Distributed Worker Sandbox

PLANNED • REVIEW 3

Redis BullMQ queue dispatching submission jobs to disposable Docker containers under gVisor security fencing to enforce 256MB/1000ms limits.

Target: Node.js worker daemon & Docker Engine

Proposed System Architecture

Decoupled Multi-Tier Blueprint

Tier 1: Presentation

IMPLEMENTED

Stack: React 19, TypeScript, Vite, Monaco Editor.
Provides code editing, language caching, test case execution interface, and real-time client state telemetry.

Tier 2: API Gateway

PLANNED (R3)

Stack: Express.js, JWT, Argon2id.
Reverse proxy and authentication gateway routing requests across Problem, Contest, and Submission microservices.

Tier 3: Execution

PLANNED (R3)

Stack: Redis BullMQ, Docker, gVisor.
FIFO queue worker pool executing code inside disposable containers with non-root privileges and no network access.

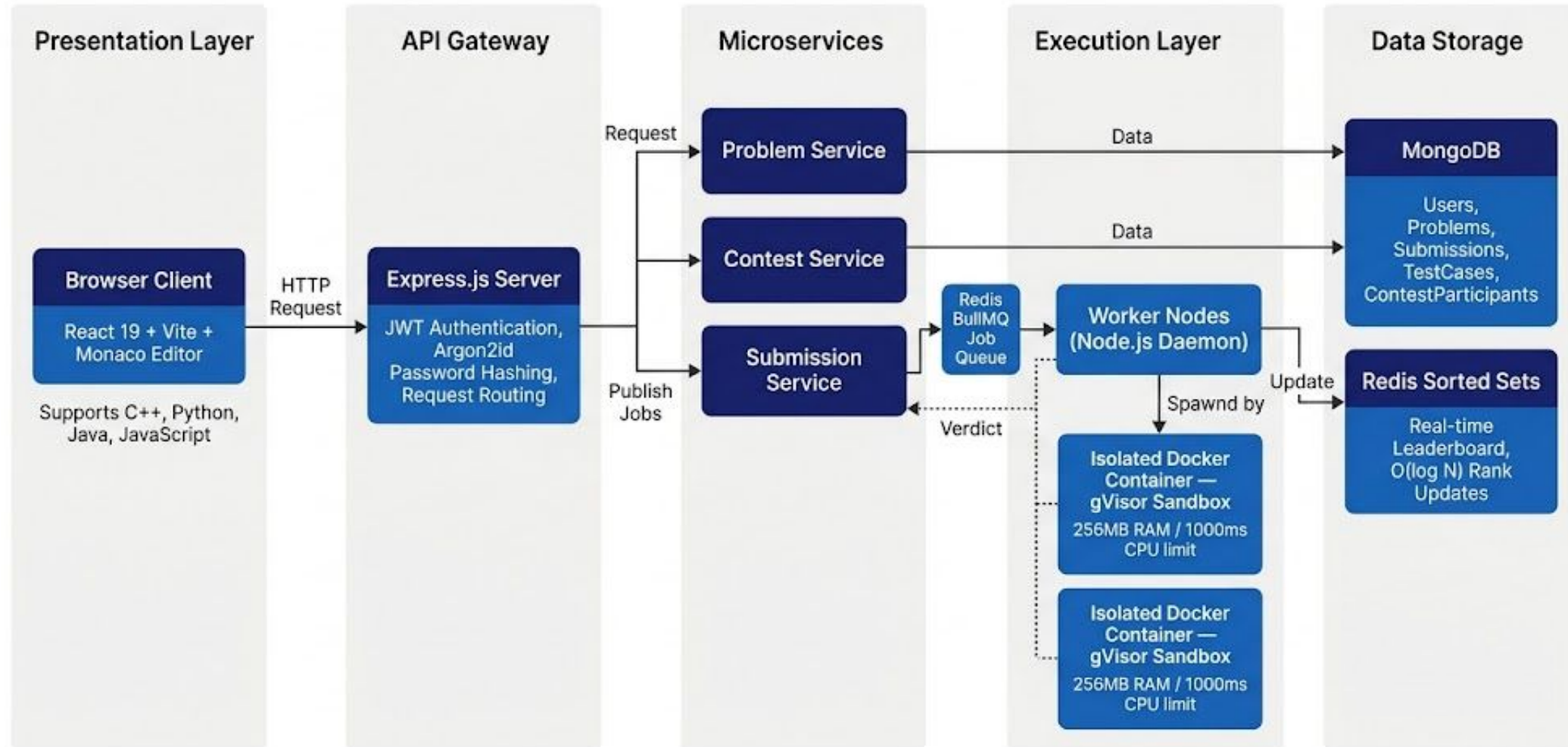
Tier 4: Persistence

DESIGNED (3NF)

Stack: MongoDB, Redis Sets.
3NF relational document storage for submissions and problems, paired with Redis Sorted Sets for $O(\log N)$ contest leaderboards.

System Architecture Diagram

Visuals



Module Implementation Status

Component-by-Component Audit

Module Name	Scope & Core Capabilities	Status	Codebase Verification
User Authentication	Sign in, sign up, and 1-click role demo profiles (Alex, Sarah, Marcus)	IMPLEMENTED	src/components/auth/AuthModal.tsx
Monaco Code Editor	Multi-language syntax highlighting, font scaling, local draft caching	IMPLEMENTED	src/components/workspace/MonacoCodeEditor.tsx
Problem Catalog	Faceted search, difficulty filtering (Easy/Med/Hard), topic tagging	IMPLEMENTED	src/components/problems/ProblemCatalog.tsx
Problem Authoring (CRUD)	Setter modal for challenge creation, resource limits, and test cases	IMPLEMENTED	src/components/problems/CreateProblemModal.tsx
Judge Simulation Engine	600ms/1400ms container latency, CPU/memory stats, 7 verdicts	IMPLEMENTED	src/context/JudgeContext.tsx
Contests & Leaderboard	Tournament round schedules with real-time ICPC penalty scoring	IMPLEMENTED	src/components/contests/LeaderboardView.tsx
Database Relational Model	3NF entity relationships, primary/foreign keys, schema contracts	DESIGN DONE	src/types/judge.ts
Distributed Worker Sandbox	Docker container execution pool under gVisor security sandbox	PLANNED (R3)	Scheduled for Phase 3 Sprint

Frontend Layout

Home

<>

AlgoFlow

Home

Problems

Contests

Leaderboard

Submissions

Search problems...

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AL

alex_dev

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Distributed Judging Engine v2.1 • Multi-Tenant Sandbox

High-Throughput Distributed Code Judge & Assessment Platform

AlgoFlow executes user-submitted code in container-isolated sandboxes with Redis BullMQ job queues, real-time telemetry, and microsecond testcase evaluation across C++, Python 3, Java, and JavaScript.

Solve Daily Challenge →

Explore 8 Problems

🏆 Live Tournaments

WORKER SANDBOXES

8 Active

gVisor Container Isolation

QUEUE LATENCY

84ms

Redis BullMQ Job Broker

EVALUATIONS

12,842+

Multi-Language Submissions

PERSISTENCE

99.98%

MongoDB Replica Set

Frontend Layout

Problem Catalogue

<>

AlgoFlow

Home

Problems

Contests

Leaderboard

Submissions

Search problems...

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🔔

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Problems

8 available

Solved: 4/8

2E • 1M • 1H

+ Create Problem

Search problems by title, topic, or ID...

All Easy Medium Hard

All Solved Unsolved

Topics: All Array Binary Search Breadth-First Search Depth-First Search Design Divide and Conquer Doubly-Linked List Dynamic Programming Graph Hash Table Linked List

STATUS	Title ↑↓	Acceptance ↑↓	Difficulty ↑↓	TOPICS
✔	1. Two Sum	51.4%	Easy	Array Hash Table
○	2. Add Two Numbers	42.1%	Medium	Linked List Math +1
✔	3. Longest Substring Without Repeating Characters	34.7%	Medium	Hash Table String +1
○	4. Median of Two Sorted Arrays	38.6%	Hard	Array Binary Search +1
✔	5. Trapping Rain Water	61.2%	Hard	Array Two Pointers +2
○	6. LRU Cache	42.8%	Medium	Hash Table Linked List +2
○	7. Course Schedule	46.9%	Medium	Depth-First Search Breadth-First Search +2

Frontend Layout

WorkSpace

< Problems / #1. Two Sum

< >

Description

Submissions 1

Editorial

1000ms • 128MB

Two Sum Easy

AlgoFlow Editorial • Acceptance: 51.4% • Submissions: 489,210

Given an array of integers `nums` and an integer `target`, return *indices* of the two numbers such that they add up to `target`*.

You may assume that each input would have **exactly one solution**, and you may not use the same element twice.

You can return the answer in any order.

SAMPLE TEST CASES

Example 1

Input: `nums = [2, 7, 11, 15], target = 9`

Output: `[0, 1]`

C++ (GCC 13.2)

13px

🔄

📄

↗

```
1  #include <vector>
2  #include <unordered_map>
3
4  class Solution {
5  public:
6      std::vector<int> twoSum(std::vector<int>& nums, int target) {
7          std::unordered_map<int, int> numMap;
8          for (int i = 0; i < nums.size(); ++i) {
9              int complement = target - nums[i];
```

> Cases:

Case 1

Case 2

Case 3

+ Custom

Input

nums = [2, 7, 11, 15], target = 9

Expected Output

[0, 1]

Actual Output

(Click "Run" to test)

Console

Sandbox Ready

Run Ctrl+R

Submit Ctrl+S

Frontend Layout

Console Runner

> Cases: **Case 1** Case 2 Case 3 + Custom

Input

```
nums = [2, 7, 11, 15], target = 9
```

Expected Output

```
[0, 1]
```

Actual Output

(Click "Run" to test)

● Sandbox Ready

▶ Run **Ctrl**

🚀 Submit **Ctrl**

Frontend Layout

Contest Area

Contests & Competitions

Live Round Active

Global Leaderboard →

Scheduled rated tournaments with real-time scoring, live rankings, and automated test penalty evaluation.

LIVE CONTEST ARENA

LIVE NOW • Division 1

Global CodeSprint 2026

Enter Contest Arena →

Started 11:30:00 PM • Concludes 2:30:00 AM

⌚ Duration: 180m

👤 Contestants: 8,840

🛡️ Integrity Guard: Active

🕒 Problems: 4

UPCOMING TOURNAMENTS

Rated (Div. 1 + Div. 2)

9/13/2026

Weekly Contest 412

Duration: 90 mins • 14,209 Registered

Scheduled in 4 days

Register Now

PAST CONTEST ARCHIVES

Tournament Round	Date	Participants	Problems	Standings
Weekly Contest 412	9/13/2026	14,209	4 Problems	Rankings

Frontend Layout

LeaderBoard

←

Global CodeSprint 2026

Official Ladder

Real-time ranked tournament scoreboard calculated with standard ICPC/competitive penalty rules.

Connection: Live WebSocket

RANK	CONTESTANT	SCORE	PENALTY	P1 (TWO SUM)	P2 (ADD TWO)	P3 (COURSE SCHED)	P4 (MEDIAN 2)
1	TO Gennady Korotkevich @tourist ITMO • Rating: 3845	400	28m	+1 3m	+1 7m	+1 11m	+1 21m
2	EC Andrew He @ecnerwala MIT • Rating: 3620	400	34m	+1 4m	+1 9m	+1 14m	+1 24m
3	AL Alex Chen @alex_dev You San Francisco, CA • Rating: 1842	300	42m	+1 5m	+1 16m	+2 31m	-1
#4	SA Sarah Kim @sarah_k Seattle, WA • Rating: 1910	300	54m	+1 6m	+2 22m	+1 38m	-
#5	MA Marcus Vance @marcus_v Austin, TX • Rating: 1725	200	29m	+1 8m	+1 21m	-2	-

Frontend Layout

Submissions

Submissions Log

7 recorded

All Verdicts ▼

ALL LANGUAGES ▼

Execution audit trail from all practice submissions and contest rounds.

PROBLEM	VERDICT	LANGUAGE	RUNTIME	MEMORY	TIME	ACTION
Two Sum	✔ Accepted	CPP	4 ms	13.8 MB	12:02 AM	Inspect
Trapping Rain Water	✔ Accepted	PYTHON	52 ms	18.5 MB	10:44 PM	Inspect
Trapping Rain Water	🕒 Time Limit Exceeded	PYTHON	1205 ms	16.0 MB	10:25 PM	Inspect
Longest Substring Without Repeating Characters	✔ Accepted	JAVASCRIPT	68 ms	45.1 MB	01:40 AM	Inspect
Valid Parentheses	✔ Accepted	CPP	0 ms	8.1 MB	07:52 PM	Inspect
LRU Cache	✖ Wrong Answer	JAVA	82 ms	66.8 MB	05:15 PM	Inspect
Course Schedule	⚠ Runtime Error	PYTHON	14 ms	16.8 MB	02:42 PM	Inspect

Frontend Layout

Dashboard

AL

Alex Chen

@alex_dev

Candidate

📍 San Francisco, CA

✉ alex.chen@example.com

CONTEST RATING

1842

Top 1.4%

GLOBAL RANK

#142

142k tracked

STREAK

🔥 28d

Active

Solved Challenges Breakdown

Verified solutions across difficulty thresholds

148 / 2,400+ total

Easy

74 / 150

49%

Medium

58 / 150

39%

Hard

16 / 50

32%

Evaluation Accuracy & Benchmark

Overall submission success and execution standards

ACCEPTANCE RATE

80%

4 of 5 attempts

DIVISION STANDING

Div 1

Rating > 1800 Tier

✅ Standardized Sandbox Guarantees

All submissions run inside isolated gVisor containers with strict CPU time quotas and memory ceiling enforcement.

Annual Activity Heatmap

947 sandbox evaluations recorded across 52 weeks

Less

🟡 🟢 🟣

 More

Hover over any square to inspect daily submission volume

Database Design & ER Modeling

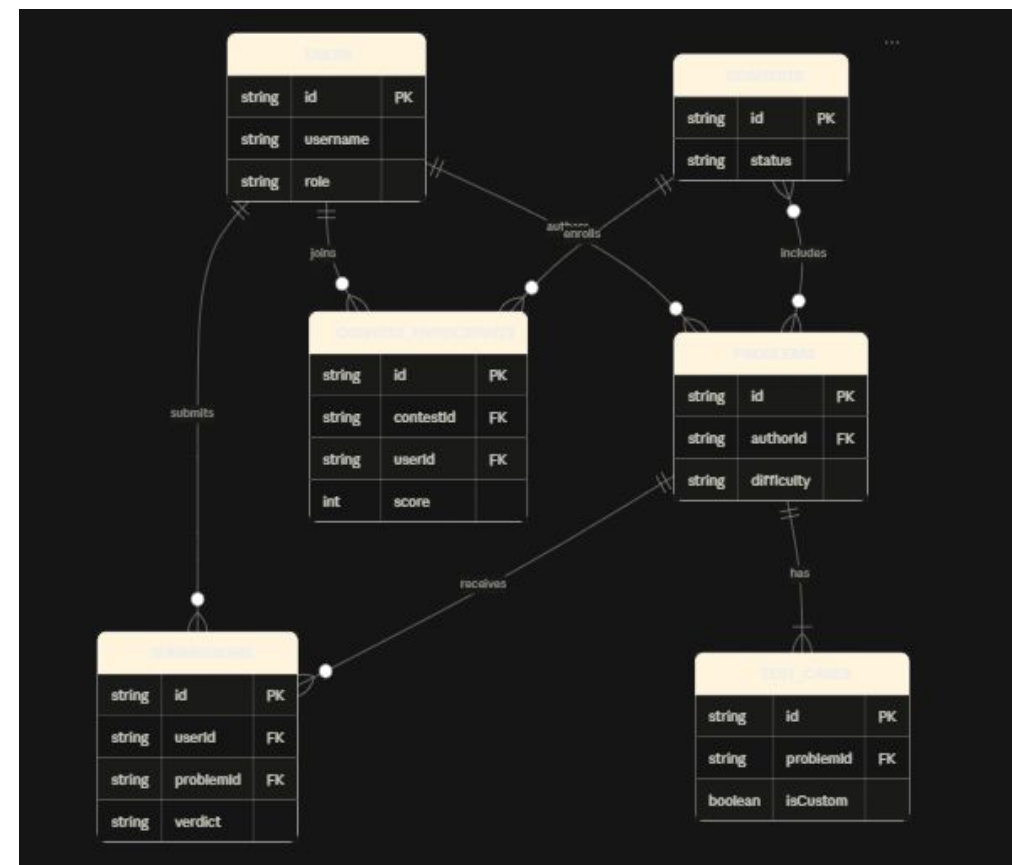
3NF Relational Structure & Cardinalities

Normalization & Rationale

3rd Normal Form (3NF) Compliance: Submissions are completely decoupled from Problems and Users, eliminating transitive dependencies and preventing table locking during write bursts.

Referential Integrity: Strict primary/foreign key relationships between Users, Submissions, Problems, and Test Cases.

Associative Entity: CONTEST_PARTICIPANT resolves the M:N relationship between Users and Contests, tracking penalty minutes and live scores.



Sample Database Records

Active Prototype Seed Datasets

 Prototype Seed Data: Populated in client-side memory stores to validate schema bindings prior to Review 3 physical database connection.

Problem ID	Title	Difficulty	Timeout	RAM Limit	Tags
prob-1	Two Sum	Easy	1000 ms	256 MB	Array, Hash Table
prob-4	Median of Two Sorted Arrays	Hard	2000 ms	256 MB	Binary Search, Divide & Conquer
prob-6	LRU Cache	Medium	1500 ms	512 MB	Linked List, Design

Sub ID	User Handle	Problem	Language	Verdict	CPU Time	Memory
sub_101	alex_dev	Two Sum	C++	Accepted	42 ms	14.2 MB
sub_102	sarah_k	Median Arrays	Python	Time Limit Exceeded	2010 ms	28.6 MB
sub_103	marcus_v	LRU Cache	Java	Wrong Answer	118 ms	32.1 MB

Challenges, Roadmap & Live Demo

Review 2 Summary & Next Steps

Challenges Solved

Editor Drag Performance: Clamped workspace math (25%–75%) to eliminate re-render stutter.

Multi-Language Buffers: Preserved unsaved contestant code drafts across language switches.

3NF Decoupling: Normalized schemas to prevent high-write submission bottlenecks.

Future

Express REST Gateway: Endpoints for judge, problems, and auth with JWT.

Live MongoDB Connection: Mongoose models reflecting normalized 3NF schemas.

Docker Sandbox Daemon: gVisor container workers consuming Redis BullMQ jobs.

Conclusion

- Successfully designed and implemented a distributed Online Code Judge frontend with Monaco Editor supporting C++, Python, Java, and JavaScript
- Completed 6 of 8 planned modules — User Auth, Code Workspace, Problem Catalog, Problem Authoring, Contests & Leaderboard, and Judge Simulation Engine
- 3NF-compliant database schema designed with full referential integrity across Users, Problems, Submissions, TestCases, and ContestParticipant
- Asynchronous sandboxed execution architecture planned via Redis BullMQ + Docker/gVisor — decoupling frontend ingestion from backend worker execution
- Review 3 targets live MongoDB integration, Express REST Gateway, and Docker Sandbox Daemon deployment